Solution:

Here, we have to select 10 random numbers ranging from 1550 to 8000 but the given random number table has only 2 digit numbers. To solve this, two - 2 digit numbers can be combined together to make a four-figure number. Let us select the 5th and 6th column and combine them to form a random number, then select the random number with given range. This gives 5 random numbers, similarly 8th and 9th is selected and combined to form a random numbers, then select the random number with given range. This gives 5 random numbers, totally 10 four-figure numbers have been selected. The following table shows the 10 random numbers which are combined and selected.

23	15	75	48	59	01	83	72	59	93	76	24	97	08	86	95	23	03	67	44
05	54	55	50	43	10	53	74	35	08	90	61	18	37	44	10	96	22	13	43
14	87	16	03	50	32	40	43	62	23	50	05	10	03	22	11	54	36	08	34
38	97	67	49	51	94	05	17	58	53	78	80	59	01	94	32	42	87	16	95
97	31	26	17	18	99	75	53	08	70	94	25	12	58	41	54	88	21	05	13

Therefore the selected 10 random numbers are

5901	4310	5032	5194	1899
7259	7435	4362	1758	5308

(ii) Stratified Random Sampling

Definition 8.1

In stratified random sampling, first divide the population into sub-populations, which are called strata. Then, the samples are selected from each of the strata through random techniques.The collection of all the samples from all strata gives the stratified random samples.

When the population is heterogeneous or different segments or groups with respect to the variable or characteristic under study, then Stratified Random Sampling method is studied. First, the population is divided into homogeneous number of sub-groups or strata before the sample is drawn. A sample is drawn from each stratum at random. Following steps are involved for selecting a random sample in a stratified random sampling method.

(a) The population is divided into different classes so that each stratum will consist of more or less homogeneous elements. The strata are so designed that they do not overlap each other.

(b) After the population is stratified, a sample of a specified size is drawn at random from each stratum using Lottery Method or Table of Random Number Method.

Stratified random sampling is applied in the field of the different legislative areas as strata in election polling, division of districts (strata) in a state etc…

Example 8.5

From the following data, select 68 random samples from the populationof heterogeneous group with size of 500 through stratified random sampling, considering the following categories as strata.

Category1: Lower income class -39%

Category2: Middle income class - 38%

Category3: Upper income class- 23%

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